Articulatory suppression effects on verbal rumination

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Theoretical background

What is rumination ?

Rumination involves **repetitive thoughts** about one's self, concerns and experiences [1], in a predominantly **verbal** form.

While rumination is a frequent process in the general population, it nonetheless has several detrimental consequences. Rumination has been shown to contribute to the development and the maintenance of more than **thirteen** disorders [2].

Inner speech

Inner speech, or silent speech, is the activity of silently talking to oneself, and plays a central role in self-awareness, memory, planning, problem-solving, reading or writing [3].

However, it can play a more negative role (e.g., auditory verbal hallucinations or rumination).

Whether inner speech consists of abstract representations or involves concrete **motor simulation** is still debated.

Articulatory suppression

Articulatory suppression consists in asking to repeat irrelevant sounds or making irrelevant movements with the speech effectors. It has been show to impair working memory performance.

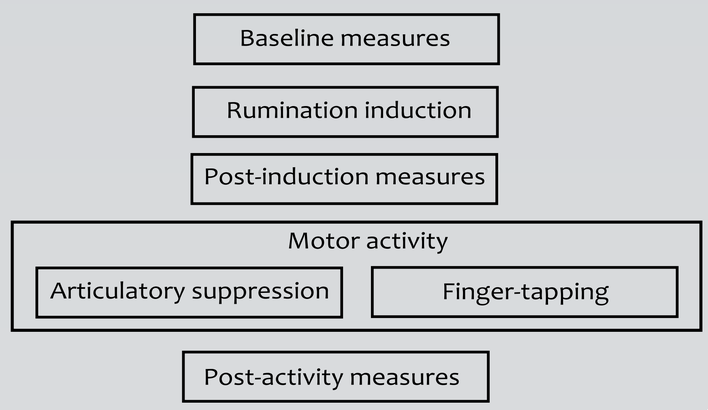
Articulatory suppression or attentionnal distraction ?

In order to demonstrate that the two motor tasks used in the main experiment (i.e., finger-tapping and mouthing) were roughly equivalent in terms of task difficulty or general **dual-task demand**, we carried out a **control experiment** with 24 participants. Participants performed a visual search task (i.e., finding a T among an array of Ls, adapted from [4]) while either finger-tapping, or mouthing or without any dual-task (control).

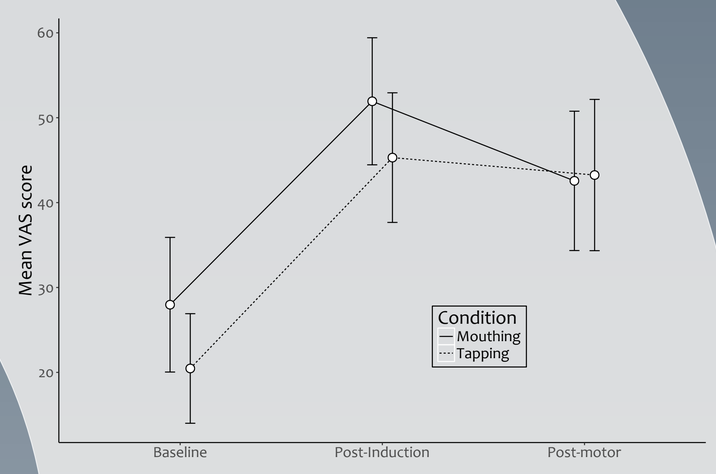
No differences were observed between the two dual-task conditions, in terms of reaction times.

Procedure

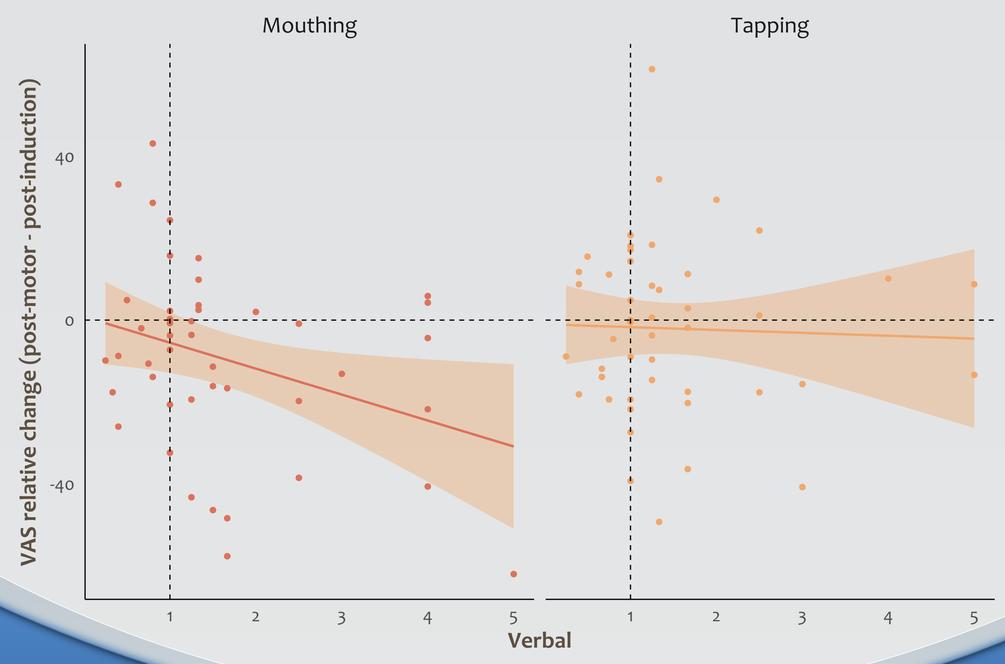
Based on a priori power analysis, we planned for 128 participants. So far, we recruited 161 participants, removed 25 with high levels of depressive symptoms, and 44 not sensitive to rumination induction. Temporary final sample, n = 92.



Results



"At this moment, I am brooding about negative things"



Reproducible research

Data, reproducible code for data analysis and figures will be made available at Open Science Framework. This study has been pre-registered.

OPEN DATA
OPEN MATERIALS
PREREGISTERED

Discussion and research directions

Articulatory supression does reduce rumination... when rumination is verbal !

We found an interaction between condition (articulatory suppression versus finger-tapping) and verbal / non-verbal ratio. In the articulatory suppression, the higher the verbal quality of rumination, the greater the decrease in self-reported rumination after the motor activity. These results suggest that verbal rumination (as an instance of inner speech) could be reduced by articulatory suppression.

Is articulatory suppression muscle specific ?

Our results seem to favour the motor simulation view of inner speech production. A further argument could come from the study of the specificity of these effects, by showing that effector-specific articulatory suppression disrupts inner production of sounds requiring the mobilized effectors (e.g., labial sounds for lip movements, versus alveolar sounds for tongue tip movements).

References

[1] Martin, L. L., & Tesser, A. (1996). Some ruminative thoughts.

[2] Baeyens, C., Kornacka, M., & Douilliez, C. (2012). De la dépression à une perspective transdiagnostique: conceptualisation et interventions sur les ruminations mentales.

[3] Perrone-Bertolotti, M., Rapin, L., Lachaux, J.-P., Baciu, M., & Loevenbruck, H. (2014). What is that little voice inside my head? Inner speech phenomenology, its role in cognitive performance, and its relation to self-monitoring.

[4] Treisman, A. M., & Gelade, G. (1980). A feature integration theory of attention.

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